

PREDICTIVE CLASSIFICATION USING RANDOM FOREST ENSEMBLES

Project 4 Report | Titanic Passenger Survival Prediction | Mayank L. Surana

1. Project Objective

The objective is to build a machine-learning classification model that predicts whether a Titanic passenger survived. The project follows the workflow of inspecting and cleaning data, exploring patterns, preparing features, training a Random Forest model, and evaluating its predictions.

2. Dataset Description

The Titanic dataset contains passenger-level records associated with the 1912 Titanic voyage. The target column, **Survived**, indicates the outcome: 0 means the passenger did not survive and 1 means the passenger survived. The dataset includes fields such as passenger class (**Pclass**), sex, age, number of siblings or spouses aboard (**SibSp**), number of parents or children aboard (**Parch**), ticket, fare, cabin, and port of embarkation (**Embarked**). These fields provide information the model can use to learn patterns associated with survival; they do not, by themselves, establish causation.

3. Data Inspection and Exploratory Data Analysis

- Inspecting sample rows with **head()**, dataset dimensions with **shape**, column types and non-null counts with **info()**, and numeric summaries with **describe()**.
- Calculating missing-value counts and percentages to identify incomplete fields.
- Plotting the distribution of the survival target to examine class balance.
- Creating a correlation heatmap for numeric columns to inspect pairwise relationships.
- Comparing survival outcomes by gender and passenger class using cross-tabulations and count plots.
- Plotting the age distribution with a histogram and KDE curve.

4. Data Cleaning and Feature Preparation

Missing **Age** values are filled with the median Age, and missing **Embarked** values are filled with the most frequent category. The **Cabin** column is removed because a large proportion of its entries are missing. The **Survived** column is separated as the target variable, while the remaining columns form the input features. Categorical input columns are converted into numeric indicator columns using one-hot encoding with **drop_first=True**.

5. Train/Test Split and Random Forest Model

The prepared data is split into training and testing subsets in an 80:20 ratio. Stratification preserves the approximate survival-class proportions across the subsets, while **random_state=42** makes the split reproducible. A **RandomForestClassifier** is initialized with **n_estimators=100**, **max_depth=10**, and **random_state=42**, then trained on the training data. The trained model predicts the survival labels for the test set.

Model-Setting Rationale

The 100 decision trees combine multiple tree predictions and can provide more stable results than relying on a single tree. Limiting the maximum depth to 10 constrains tree growth and may help reduce overfitting. These are reasonable starting settings; cross-validation and parameter tuning could be used to investigate whether other settings perform better.

6. Evaluation Metrics and Visualizations

The project evaluates the model using several complementary measures. A confusion-matrix heatmap displays true negatives, false positives, false negatives, and true positives.

- **Precision**: the proportion of passengers predicted as survivors who actually survived.
- **Recall**: the proportion of actual survivors correctly identified by the model.
- **F1-score**: the harmonic mean of precision and recall.
- **ROC curve and ROC-AUC**: the curve compares true-positive and false-positive rates across thresholds; the AUC summarizes how well the model separates the two classes.
- **Feature importance**: a bar chart shows the ten features with the highest importance scores in the fitted Random Forest.

7. Feature Scaling

Feature scaling is not applied because Random Forest uses decision-tree splits based on feature thresholds. Its performance generally does not require numeric features to be standardized or normalized, unlike some distance-based algorithms.

8. Conclusion and Limitations

This project demonstrates a complete supervised-classification workflow, from exploratory analysis and preprocessing to ensemble model training and evaluation. The confusion matrix, precision, recall, F1-score, ROC-AUC, and feature-importance chart provide different perspectives on predictive performance. The exact metric values should be copied from the executed notebook output; they are not stated in this report because those results were not available during report preparation. Feature importance describes the model's use of features for prediction and should not be interpreted as proof of causation. Performance can also vary with different data splits and model settings.

Component	Implementation
Target variable	Survived (0 = did not survive; 1 = survived)
Data split	80% training / 20% testing; stratified; random_state=42
Model	RandomForestClassifier; 100 trees; maximum depth 10
Evaluation	Confusion matrix, precision, recall, F1-score, ROC-AUC, feature importance